ChatGPT said:

Lowest Common Ancestor (LCA) — Notes

1. Introduction

The **Lowest Common Ancestor (LCA)** is a concept in a tree data structure. It is the lowest (deepest) node in a tree that is an ancestor of two given nodes.

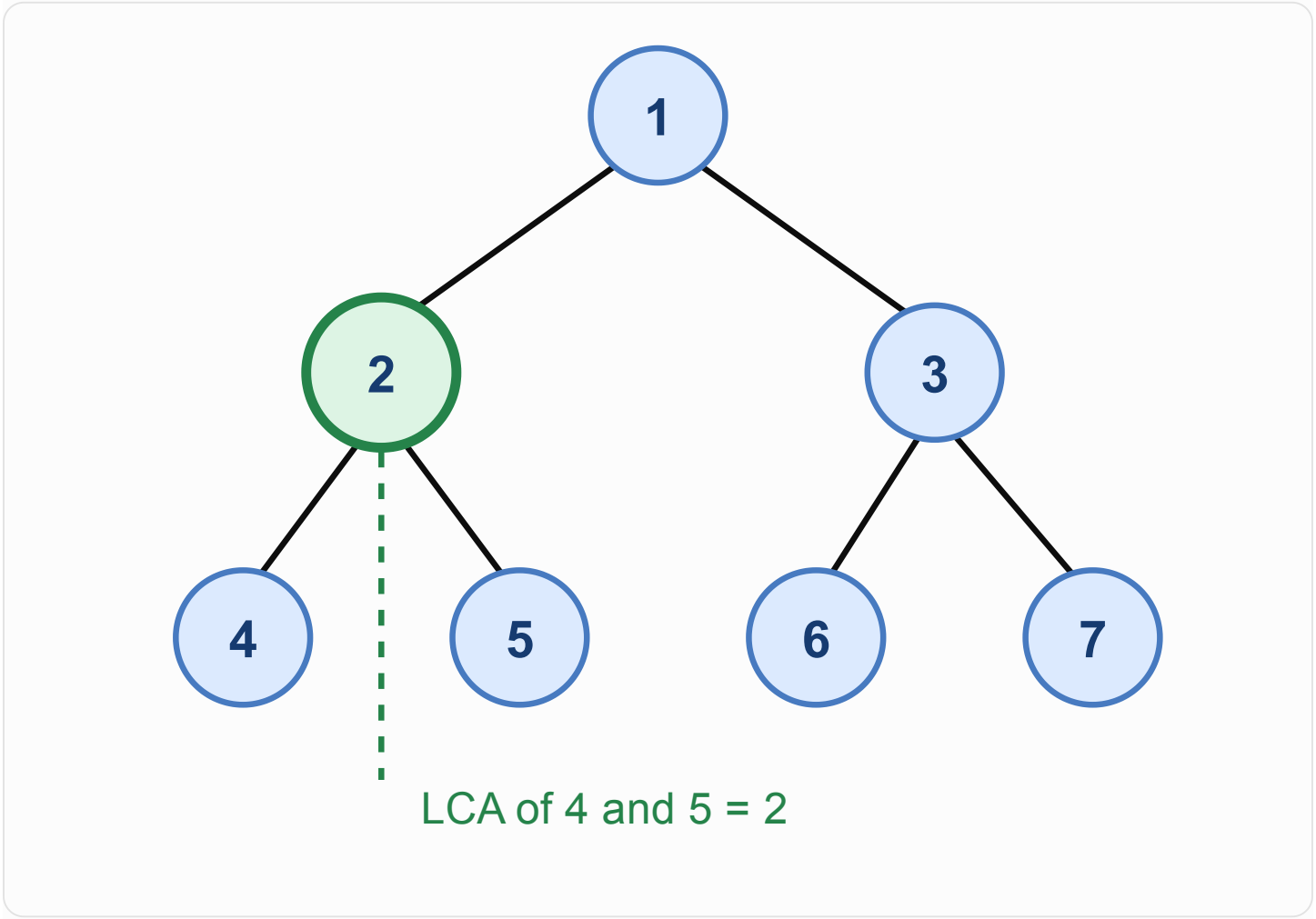
An **ancestor** of a node is any node that lies on the path from the root to that node. A node can also be considered its own ancestor.

Definition

The Lowest Common Ancestor of two nodes p and q is the deepest node that has both p and q as its descendants.

2. Example of Lowest Common Ancestor

Consider the following binary tree:



In the above tree:

- The LCA of nodes 4 and 5 is node 2.
- The LCA of nodes 4 and 6 is node 1.

- The LCA of nodes 6 and 7 is node 3.

Explanation: Nodes 4 and 5 have node 2 as their common ancestor. Since node 2 is the deepest common ancestor, it is their LCA.

3. Properties of LCA

1. The LCA is the deepest common ancestor of two nodes.
2. The root can be the LCA of two nodes.
3. If one node is an ancestor of another, the ancestor node is the LCA.
4. The LCA of a node with itself is the same node.
5. LCA is commonly used in binary trees, binary search trees, and other tree-based problems.

4. Finding LCA in a Binary Search Tree (BST)

In a Binary Search Tree:

- Values smaller than the root are present in the left subtree.
- Values greater than the root are present in the right subtree.

We can find the LCA by comparing the two given values with the current node.

Algorithm

1. Start from the root node.
2. If both values are smaller than the current node, move to the left subtree.
3. If both values are greater than the current node, move to the right subtree.
4. Otherwise, the current node is the LCA.

Example

Suppose the BST contains nodes 2, 4, 6, 8, 10, 12 and 14, with root 8.

To find the LCA of 4 and 6:

- Both values are smaller than 8, so move left.
- Node 4 is the ancestor of node 6.
- Therefore, the LCA is **node 4**.

5. Finding LCA in a General Binary Tree

In a general binary tree, values do not necessarily follow BST ordering rules.

A recursive approach is commonly used:

1. If the current node is `NULL`, return `NULL`.
2. If the current node matches either of the given nodes, return the current node.
3. Search for the nodes in the left and right subtrees.

4. If both subtrees return a node, the current node is the LCA.
5. Otherwise, return whichever subtree returned a non-`NULL` node.

6. Time and Space Complexity

Method	Time Complexity	Auxiliary Space
BST (balanced)	$O(\log n)$	$O(1)$ iterative
BST (skewed)	$O(n)$	$O(1)$ iterative
Binary tree (recursive)	$O(n)$	$O(h)$

Here, n is the number of nodes and h is the height of the tree.

7. Applications of LCA

1. Finding the relationship between two nodes in a tree.
2. Finding the distance between two nodes.
3. Solving problems involving family trees and hierarchies.
4. Network routing and path-related problems.
5. Solving tree problems in competitive programming and technical interviews.

8. Conclusion

The Lowest Common Ancestor is the deepest node shared by two nodes as their ancestor. It can be found efficiently using BST properties or recursive traversal in a general binary tree.

Important exam point: LCA is the *lowest common ancestor*, not necessarily the node with the smallest numerical value.